

CLINICAL CRITERIA FOR SUBCLINICAL INFECTION

MAJOR CRITERIA WITNESS STAND

Q FEVER WITNESS

TYPICAL ORGANISM
IN ≥ 2 SEPARATE SETS
— *S. aureus*, *E. faecalis*,
Streptococci, *HACEK*

HACEK
IN ≥ 2 BLOOD CULTURE
SETS = MAJOR
— OROM
— NO NEED FOR 3RD
CULTURE IF ALREADY
DEFINITE IE BY CRITERIA

ORAL CAVITY ORIGIN

MAJOR IMAGING CRITERIA

ECHO/CT:
VEGETATION
— PERFORATION
— ABSCESS
— FISTULA
— NEW VALVULAR
REGURGITATION
ON ECHO = MAJOR

**NEW MURMUR ON
AUSCULTATION ONLY**
= MINOR CRITERION G
— NOT MAJOR —
— NEEDS ECHO
CONFIRMATION

MAJOR CRITERIA

2 MAJOR =
DEFINITE IE

1 MAJOR +
3 MINOR =
DEFINITE IE

5 MINOR =
DEFINITE IE

1 MAJOR +
1 MINOR =
POSSIBLE IE

3 MINOR =
POSSIBLE IE

C. BURNETII
PHASE I
IgG $\geq 1:800$
= MAJOR
CRITERION

BARTONELLA
IgG $\geq 1:800$
= MAJOR

CATS =
— *B. HENSELAE*
— BODY LICE
— *B. QUINTANA*

MINOR CRITERIA GALLERY

PREDISPOSITION
— PROSTHETIC VALVE
— PRIOR IE / IVDU
/ CHD / HCM

FEVER $>38^{\circ}\text{C}$

IMMUNOLOGIC
— OSLER NODES
— ROTH SPOTS
— GLOMERULONEPHRITIS
— POSITIVE RF

OSLER = PAINFUL
— IMMUNOLOGIC
— JANEWAY = PAINLESS
— VASCULAR

PETECCHIAEC
— FONELS
— PEREIN ECGN
— POSITIVE RF

VASCULAR PHENOMENA
— JANEWAY LESIONS
— SEPTIC EMBOLI
— MYCOTIC ANEURYSM
— CONJUNCTIVAL HEMORRHAGE

POSITIVE CULTURES
NOT MEETING
MAJOR CRITERIA

**NEW MURMUR ON
AUSCULTATION**
— MINOR ONLY —
— NO ECHO CONFIRMATION

TEE vs TTE COMPARISON BOOTH

TTE —
— MISSES 20–35%
— CANNOT DETECT $<2\text{mm}$
— USE FOR LOW RISK /
LOW SUSPICION
— FIRST STEP

TEE —
— $>90\%$ SENSITIVITY
— TEST OF CHOICE FOR
HIGH RISK
— PROSTHETIC
VALVES
— ABSCESSES

**NEGATIVE TEE DOES
NOT EXCLUDE IE**
— REPEAT IN 7–10 DAYS
IF SUSPICION PERSISTS

PET-CT
— FOR PVE / TAVR / CIED
— NOT FIRST LINE
FOR NVE
— FALSE POSITIVE
WITHIN 3 MONTHS
MONTHS OF VALVE
SURGERY

Q35 CASE CARD

**PROSTHETIC VALVE (MINOR A) +
FEVER (MINOR B) +
NEW MURMUR AUSCULTATION (MINOR G)**
= 3 MINOR = POSSIBLE IE
→ ADD POSITIVE BLOOD CULTURE
= 1 MAJOR + 3 MINOR = DEFINITE IE

TEE vs TTE COMPARISON BOOTH

1. Typical organism — major

WHAT YOU SEE

Witness stand: typical organism placard

WHAT IT ENCODES

A major blood-culture criterion is a typical IE organism (Viridans strep, *S. bovis/gallolyticus*, HACEK, *S. aureus*, community enterococcus without primary focus) from 2 separate cultures.

BOARD TRAP

S. aureus counts as a major criterion even from community sources here — don't dismiss it as a 'contaminant'.

2. Persistent bacteremia rule

WHAT YOU SEE

2 separate sets / sustained positivity

WHAT IT ENCODES

Persistently positive cultures (2 sets >12 h apart, or 3/4+ sets with first and last >1 h apart) is a major criterion even for less-typical organisms, reflecting endovascular infection.

BOARD TRAP

A single positive culture (except *Coxiella*) is NOT a major criterion — needs the temporal/multiple-set pattern.

3. Q fever / Coxiella

WHAT YOU SEE

Q FEVER WITNESS / *C. burnetii* phase I IgG ≥1:800

WHAT IT ENCODES

A single positive *Coxiella burnetii* culture OR phase I IgG titer ≥1:800 is by itself a major criterion. Q fever IE is culture-negative and needs serology to diagnose.

BOARD TRAP

Coxiella is the exception where ONE positive serology/culture suffices — don't require two sets.

4. Bartonella

WHAT YOU SEE

Bartonella IgG ≥1:800 / cats

WHAT IT ENCODES

Bartonella (*henselae/quintana*) is a culture-negative IE cause; serology IgG ≥1:800 supports it. Associated with cats (*henselae*) and homeless/body-louse exposure (*quintana*).

BOARD TRAP

Culture-negative IE is not 'no IE' — think *Coxiella*, *Bartonella*, HACEK, *Tropheryma* before excluding.

5. HACEK organisms

WHAT YOU SEE

HACEK in 2 cultures = major

WHAT IT ENCODES

HACEK (*Haemophilus*, *Aggregatibacter*, *Cardiobacterium*, *Eikenella*, *Kingella*) are fastidious gram-negatives; modern systems incubate long enough that prolonged hold is usually unneeded. Count as typical organisms.

BOARD TRAP

HACEK no longer routinely requires extended incubation on automated systems — old teaching.

6. Predisposition (minor)

WHAT YOU SEE

Minor gallery: predisposition / prosthetic valve / IVDU

WHAT IT ENCODES

Minor criterion: predisposing heart condition (prosthetic valve, prior IE, congenital lesion) OR injection drug use. Sets the pretest probability for the rest of Duke scoring.

BOARD TRAP

Predisposition is only a MINOR criterion — it never confirms IE by itself.

7. Fever >38°C (minor)

WHAT YOU SEE

Thermometer >38°C

WHAT IT ENCODES

Fever ≥38°C is a minor criterion. Nearly universal but nonspecific; combined with vascular/immunologic phenomena and a single positive culture it builds toward 'possible' or 'definite' IE.

BOARD TRAP

Fever alone, even with predisposition, cannot reach 'definite' — need major criteria or enough minors.

8. Immunologic phenomena

WHAT YOU SEE

Osler nodes, Roth spots, GN, RF

WHAT IT ENCODES

Minor immunologic criterion: Osler nodes (painful, immune), Roth spots, glomerulonephritis, positive rheumatoid factor. Reflect immune-complex deposition in subacute IE.

BOARD TRAP

Osler = Ouch (painful, immunologic); Janeway = painless (embolic, vascular) — classic swap trap.

9. Vascular phenomena

WHAT YOU SEE

Janeway lesions, septic emboli, mycotic aneurysm, conjunctival/ICH

WHAT IT ENCODES

Minor vascular criterion: arterial/septic emboli, septic pulmonary infarcts, mycotic aneurysm, intracranial/conjunctival hemorrhage, painless Janeway lesions.

BOARD TRAP

Janeway lesions are vascular/embolic and painLESS — do not classify them as immunologic Osler nodes.

10. Microbiologic minor

WHAT YOU SEE

Positive cultures not meeting major criteria

WHAT IT ENCODES

A positive blood culture not meeting major criteria (or serologic evidence of active infection) is a minor criterion. Lets atypical/single positives still contribute.

BOARD TRAP

A skin-flora single positive may be a contaminant — context decides if it counts as a minor.

11. New murmur (minor)

WHAT YOU SEE

New murmur on auscultation / echo confirmation

WHAT IT ENCODES

New valvular regurgitation murmur is a minor clinical clue but must be confirmed by echo to count as the imaging major criterion. Worsening of an old murmur isn't sufficient.

BOARD TRAP

Change in a pre-existing murmur does NOT count as the 'new regurgitation' imaging criterion — must be genuinely new.

12. Echo major — vegetation

WHAT YOU SEE

Major imaging: vegetation, abscess, fistula, new regurg

WHAT IT ENCODES

Echo major criterion = vegetation, abscess, pseudoaneurysm, fistula, valve perforation, new partial prosthetic dehiscence, or new valvular regurgitation.

BOARD TRAP

Worsening of pre-existing valve disease is not the criterion — must be NEW regurgitation/structural lesion.

13. Q35 case card scoring

WHAT YOU SEE

Q35 CASE CARD: 1 major + 3 minor = definite

WHAT IT ENCODES

Definite IE = 2 major, OR 1 major + 3 minor, OR 5 minor. Possible IE = 1 major + 1 minor, OR 3 minor. Pathologic confirmation also yields definite.

BOARD TRAP

Memorize the exact combos — 1 major + 1 minor is only 'possible', not 'definite'.

14. TTE first-line

WHAT YOU SEE

TTE booth: misses 20-35%, cannot detect abscess

WHAT IT ENCODES

TTE is the initial test (quick, noninvasive) but misses 20-35% of vegetations and cannot reliably detect abscess or assess prosthetic valves.

BOARD TRAP

A negative TTE does NOT exclude IE — escalate to TEE when suspicion is high.

15. TEE sensitivity

WHAT YOU SEE

TEE: >90% sensitivity, test of choice

WHAT IT ENCODES

TEE has >90% sensitivity, is the test of choice for abscess, prosthetic valves, and small vegetations. If initial TEE negative but suspicion persists, repeat in 7-10 days.

BOARD TRAP

One negative TEE early doesn't rule out IE — repeat in 7-10 days if clinical suspicion remains high.

16. PET-CT role

WHAT YOU SEE

PET-CT booth: prosthetic valve, not first-line

WHAT IT ENCODES

FDG-PET/CT helps in prosthetic-valve IE and detecting peripheral emboli; added to modified criteria for prostheses. Limited within ~3 months of surgery (post-op inflammation causes false positives).

BOARD TRAP

PET-CT is NOT first-line for native-valve IE and is unreliable in the early post-surgical window.